DataBase Lab 05 (POST-Lab Task)

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Task 11

Welcome Page | hr

Worksheet | Query Builder

```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id
HAVING COUNT(*) > 5;
```

Script Output | Query Result

SQL | All Rows Fetched: 5 in 0.001 seconds

	DEPARTMENT_ID	TOTAL_EMPLOYEES
1	100	6
2	30	6
3	50	45
4	80	34
5	60	6

Task 12

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Worksheet | Query Builder

```
SELECT department_id, AVG(salary) AS avg_salary
FROM employees
GROUP BY department_id
HAVING AVG(salary) > 60000;
```

Script Output | Query Result

SQL | All Rows Fetched: 0 in 0.002 seconds

DEPARTM...	AVG_SAL...

Task 13

Welcome Page | hr

Worksheet | Query Builder

```
SELECT EXTRACT(YEAR FROM hire_date) AS hire_year,
       COUNT(*) AS total_employees
FROM employees
GROUP BY EXTRACT(YEAR FROM hire_date)
ORDER BY hire_year;
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.002 seconds

	HIRE_YEAR	TOTAL_EMPLOYEES
1	2001	1
2	2002	7
3	2003	6
4	2004	10
5	2005	29
6	2006	24
7	2007	19
8	2008	11
9	2025	1
10	2026	1

Task 14

Welcome Page | hr

Worksheet | Query Builder

```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id
ORDER BY department_id;
```

Script Output | Query Result

SQL | All Rows Fetched: 12 in 0.002 seconds

	DEPARTMENT_ID	TOTAL_EMPLOYEES
1	10	1
2	20	2
3	30	6
4	40	1
5	50	45
6	60	6
7	70	1
8	80	34
9	90	3
10	100	6
11	110	2
12	(null)	2

Task 15

Welcome Page | hr

Worksheet | Query Builder

```
SELECT job_id, COUNT(*) AS total_employees
FROM employees
GROUP BY job_id
HAVING COUNT(*) > 3;
```

Script Output | Query Result

SQL | All Rows Fetched: 8 in 0.002 seconds

	JOB_ID	TOTAL_EMPLOYEES
1	FI_ACCOUNT	5
2	IT_PROG	6
3	PU_CLERK	6
4	SA_MAN	5
5	SA_REP	30
6	SH_CLERK	20
7	ST_CLERK	20
8	ST_MAN	5

Task 16

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Worksheet | Query Builder

```
SELECT department_id, MAX(salary) AS max_salary
FROM employees
GROUP BY department_id
HAVING MAX(salary) > 80000;
```

Script Output | Query Result

SQL | All Rows Fetched: 0 in 0.002 seconds

DEPARTM...	MAX_SAL...

Task 17

Welcome Page | hr

Worksheet | Query Builder

```
INSERT INTO job_history
VALUES (101, DATE '2022-01-01', DATE '2022-12-31', 'IT_PROG', 60);
INSERT INTO job_history
VALUES (102, DATE '2021-01-01', DATE '2021-12-31', 'SA_REP', 80);
INSERT INTO job_history
VALUES (103, DATE '2020-01-01', DATE '2020-12-31', 'HR_REP', 40);
INSERT INTO job_history
VALUES (104, DATE '2019-01-01', DATE '2019-12-31', 'MK_MAN', 20);
INSERT INTO job_history
VALUES (105, DATE '2018-01-01', DATE '2018-12-31', 'FI_ACCOUNT', 100);
```

Script Output | Task completed in 0.026 seconds

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

Task 17 (Continue)

Welcome Page | hr

Worksheet | Query Builder

```
INSERT INTO departments
VALUES (300, 'AI Research', NULL, 1700);
INSERT INTO departments
VALUES (310, 'Cyber Security', NULL, 1700);
INSERT INTO departments
VALUES (320, 'Cloud Services', NULL, 1700);
INSERT INTO departments
VALUES (330, 'Business Analytics', NULL, 1700);
INSERT INTO departments
VALUES (340, 'Game Development', NULL, 1700);
```

Script Output

Task completed in 0.027 seconds

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

Task 18

Welcome Page | hr

Worksheet | Query Builder

```
CREATE TABLE tasks AS
SELECT employee_id, first_name, last_name, department_id, salary
FROM employees
WHERE 1 = 2;
```

Script Output

Task completed in 0.085 seconds

```
Table TASKS created.
```

Task 18 (Continue)

Welcome Page | hr

Worksheet | Query Builder

```
INSERT INTO tasks (employee_id, first_name, last_name, department_id, salary)
SELECT employee_id, first_name, last_name, department_id, salary
FROM employees;
```

Script Output

Task completed in 0.024 seconds

```
Table TASKS created.


107 rows inserted.
```

Task 19

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Worksheet | Query Builder

```
UPDATE employees
SET department_id = 90
WHERE employee_id = 112;
```

Script Output | Task completed in 0.026 seconds

```
1 row updated.
```

Task 20

Welcome Page | hr

Worksheet | Query Builder

```
DELETE FROM job_history
WHERE employee_id IN (201, 202);

DELETE FROM employees
WHERE employee_id IN (201, 202);

commit;
```

Script Output | Query Result | Task completed in 0.024 seconds

```
0 rows updated.

1 row deleted.

2 rows deleted.

Commit complete.
```